English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 17.12.2022 Test 1

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (1 point) A football match can be decided by a penalty shootout. Assume the following procedure: each team takes one penalty in each of five rounds. If the scores are tied afterwards, rounds continue until one team scores and the other does not.

1. (0.25 points) What is the probability that, in the first five rounds, both teams score in exactly the same rounds?
2. (0.25 points) What is the probability of a tie after five rounds? And after ten rounds?
3. (0.5 points) What is the expected total number of penalties taken by both teams?

Problem 2. (1 point) A bag contains two types of six-sided dice labelled 1 through 6: fair dice, and dice that show six with probability 40% and every other face with probability 12%. As we have no information about their counts, we initially assume a 50% probability of drawing a fair die. To learn more, draw one die, roll it, replace it, and repeat. If both rolls show six, how would you update your beliefs about the types of dice in the bag?

Problem 3. (0.5 points) Choose X randomly on a unit segment OA. Then choose Y randomly on XA. What is the probability that OX, XY, YA can form a triangle?

Problem 4. (1 point) Ten couples attend a dance class. For an introductory game, the 20 people are randomly divided into teams A and B.

1. What is the expected number of people in A? What is the expected number of couples assigned to the same team?
2. Someone unhappy about being separated from their partner may request a new division. If a particular person requests repeated divisions until they are with their partner, how many divisions are needed on average?
3. Suppose all couples want to remain together but do not say so, instead waiting until this happens under random divisions. How many divisions are needed on average to obtain this configuration? What is the expected number of couples in A?